

RANONG, Thailand

WMO: 485320

Lat: 9.778N

Lon: 98.585E

Elev: 17

StdP: 101.12

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	21.3	22.1	16.7	11.9	26.1	17.9	12.8	26.6	6.1	28.3	5.2	29.0	0.4	60	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.7	35.4	25.6	34.6	25.6	33.9	25.6	27.4	31.8	27.1	31.5	26.8	31.1	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.4	26.0	21.4	29.2	25.7	21.0	29.0	87.1	31.9	85.5	31.8	84.3	31.4	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	4.7	4.0	DB	19.0	36.8	1.3	1.2	18.0	37.7	17.3	38.4	16.6	39.1	15.6	40.0	
			WB	17.3	28.5	1.4	0.9	16.3	29.1	15.5	29.6	14.7	30.1	13.7	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.3	28.0	28.9	29.4	28.5	27.8	27.4	27.4	27.1	27.2	27.4	27.1
	DBStd	1.35	1.02	0.91	1.13	1.10	1.41	1.25	1.21	1.22	1.16	1.05	1.04	1.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6491	536	504	586	581	573	534	541	538	513	534	522	529
	CDD18.3	3449	277	271	328	331	315	284	283	279	263	276	272	271
	CDH23.3	35147	2742	2986	3865	3933	3385	2899	2751	2686	2369	2466	2551	2514
CDH26.7	13171	1105	1395	1888	1842	1274	934	812	770	652	749	871	878	
Wind	WSAvg	1.1	1.2	1.1	1.1	1.0	1.0	1.2	1.2	1.3	1.0	0.8	0.9	1.3
Precipitation	PrecAvg	4202	23	16	74	142	511	662	709	735	683	438	150	60
	PrecMax	5378	247	77	420	686	994	1015	1356	1271	1150	1094	458	321
	PrecMin	3216	0	0	0	0	209	314	362	276	306	117	27	0
	PrecStd	618	48	20	98	138	210	201	221	240	222	219	104	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	35.2	36.5	36.5	35.6	33.3	32.6	32.2	32.6	33.0	33.3	33.2
		MCWB	24.2	24.5	25.1	25.7	26.6	26.3	26.0	25.9	26.1	25.8	25.6	25.1
	2%	DB	33.0	34.3	35.4	35.3	34.2	32.5	31.9	31.7	31.8	32.1	32.6	32.4
		MCWB	24.3	24.6	25.2	26.2	26.7	26.2	26.1	25.9	25.9	25.8	25.5	24.8
	5%	DB	32.3	33.6	34.6	34.4	33.0	31.8	31.2	31.1	31.0	31.4	31.9	31.7
		MCWB	24.3	24.5	25.2	26.3	26.5	26.1	26.0	25.9	25.9	25.8	25.5	24.6
	10%	DB	31.6	32.8	33.7	33.6	32.0	31.0	30.4	30.4	30.1	30.5	31.1	30.9
		MCWB	24.2	24.4	25.2	26.2	26.4	26.1	25.9	25.9	25.7	25.7	25.3	24.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.6	27.1	27.9	27.9	27.5	27.2	27.2	27.0	27.2	27.0	26.7
		MCDB	31.2	32.3	33.1	33.1	32.5	31.0	30.2	30.0	30.6	30.8	31.2	30.8
	2%	WB	25.7	25.9	26.6	27.2	27.3	27.0	26.7	26.6	26.5	26.6	26.4	26.0
		MCDB	30.7	32.0	32.6	32.7	31.9	30.6	30.0	29.8	30.1	30.3	30.8	30.5
	5%	WB	25.1	25.5	26.2	26.9	27.0	26.6	26.4	26.2	26.1	26.2	26.0	25.5
		MCDB	30.4	31.6	32.2	32.5	31.4	30.2	29.7	29.5	29.7	29.9	30.5	30.0
	10%	WB	24.7	25.0	25.8	26.6	26.6	26.2	26.1	26.0	25.9	25.9	25.6	25.0
		MCDB	29.9	31.2	31.7	31.9	30.7	29.7	29.4	29.4	29.2	29.5	29.8	29.4
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.1	9.9	8.7	6.6	5.7	5.5	5.3	5.6	6.3	7.0	7.7
		MCDBR	10.2	11.1	11.3	9.9	8.2	7.1	6.6	6.6	7.0	8.0	8.8	9.2
		MCWBR	3.6	3.7	3.6	3.1	2.8	2.5	2.4	2.4	2.6	3.2	3.3	3.4
	5% WB	MCDBR	9.1	10.1	10.0	9.0	7.3	6.3	5.9	5.8	6.3	7.2	7.8	8.1
		MCWBR	3.7	3.9	3.7	3.3	2.8	2.7	2.6	2.7	2.7	3.2	3.3	3.4
Clear-Sky Solar Irradiance	taub	0.482	0.498	0.506	0.480	0.457	0.459	0.460	0.451	0.434	0.453	0.475	0.481	
	taud	2.200	2.146	2.169	2.249	2.358	2.379	2.360	2.380	2.422	2.349	2.263	2.215	
	Ebn at Noon	828	829	826	838	841	829	832	851	876	859	829	819	
	Edh at Noon	145	158	156	143	125	121	124	124	120	127	135	141	
All-Sky Solar Radiation	RadAvg	5.00	5.82	5.97	5.89	4.71	4.42	4.26	4.38	4.44	4.34	4.38	4.38	
	RadStd	0.43	0.34	0.50	0.38	0.43	0.31	0.40	0.43	0.43	0.46	0.39	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.17	+0.42	N/A	N/A	N/A	+0.20	N/A	N/A	N/A	N/A

Nomenclature: See separate page